ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/354166915

Molecular dynamics simulation of Toxin–Antitoxin (TA) system in Acinetobacter baumannii to explore the novel mechanism for inhibition of cell wall biosynthesis: Zeta Toxin as an ef…

Article *in* Journal of Cellular Biochemistry · August 2021

DOI: 10.1002/jcb.30137

CITATIONS
0

READS
74

7 authors, including:



Karthika Alagesan
Alagappa University
1 PUBLICATION **0** CITATIONS
SEE PROFILE



Dhamodharan Prabhu
Sree Balaji Medical College and Hospital
30 PUBLICATIONS **178** CITATIONS
SEE PROFILE



Rajamanikandan Sundarraj
Karpagam University
30 PUBLICATIONS **244** CITATIONS
SEE PROFILE

Some of the authors of this publication are also working on these related projects:

Structural and functional insights of potential anti-malarial drug targets of G6PD and 6PGD from Plasmodium falciparum (3D7) View project

Structural studies on SARS CoV 2 targets View project

All content following this page was uploaded by Karthika Alagesan on 28 September 2021.

The user has requested enhancement of the downloaded file.

Check for updates

Received: 4 May 2021 | Revised: 8 August 2021 | Accepted: 10 August 2021

DOI: 10.1002/jcb.30137

RESEARCH ARTICLE

Journal of Cellular Biochemistry WILEY

Molecular dynamics simulation of Toxin-Antitoxin (TA) system in *Acinetobacter baumannii* to explore the novel mechanism for inhibition of cell wall biosynthesis: Zeta Toxin as an effective therapeutic target

Alagesan Karthika[1] | **Balajee Ramachandran**[1] | **Jeyarajpandian Chitra**[2] | **Dhamodharan Prabhu**[1] | **Sundaraj Rajamanikandan**[1] | **Malaisamy Veerapandiyan**[1] | **Jeyaraman Jeyakanthan**[1]

[1]Structural Biology and Bio-Computing Lab, Department of Bioinformatics, Alagappa University, Karaikudi, Tamil Nadu, India

[2]Department of Biotechnology, Dr. Umayal Ramanathan College for Women, Karaikudi, Tamil Nadu, India

Correspondence
Jeyaraman Jeyakanthan, Structural Biology and Bio-Computing Lab, Department of Bioinformatics, Alagappa University, Karaikudi 630 004, Tamil Nadu, India.
Email: jjkanthan@gmail.com or jjeyakanthan@alagappauniversity.ac.in

Funding information

DST – Promotion of University Research and Scientific Excellence (PURSE), Grant/Award Numbers: No. SR/PURSE Phase2/38) (G), 2017; UGC-RA-2016-18-OB-AM-7124 dated: 18.04.2016; DST-SERB, Grant/Award Number: File No. EMR/2016/000498 dated: 26.09.2016; ICMR, Grant/Award Number: Sanction Order Ref. No. (BIC/12(07)/2015 dated: 28.07.2017; MHRD-RUSA 2.0, New Delhi, Grant/Award Numbers: F.24-51/2014-U, Policy (TNMulti-Gen), Dept. of Edn. Govt. of India dated: 09.10.2018; DST –FIST, Grant/Award Number: SR/FST/LSI-667/2016; DST Indo-Taiwan project, Grant/Award Number: GITA/DST/TWN/P-86/2019 dated: 04/03/2020

Abstract

The majority of bacteria and archaea contains Toxin-Antitoxin system (TA) that codes for the stable Toxin and unstable Antitoxin components forming a complex. The Antitoxin inhibits the catalytic activities of the Toxin. In general, the Antitoxin will be degraded by the proteases leading to the Toxin activation that subsequently targets essential cellular processes, including transcription, translation, replication, cell division, and cell wall biosynthesis. The Zeta Toxin-Epsilon Antitoxin system in ESKAPE pathogen stabilizes the resistance plasmid and promotes pathogenicity. The known TA system in *Acinetobacter baumannii* are known to be involved in the replication and translation, however, the mechanism of Zeta Toxin-Epsilon Antitoxin in cell wall biosynthesis remains unknown. In the present study, molecular docking and molecular dynamic (MD) simulations were employed to demonstrate whether Zeta Toxin can impair cell wall synthesis in *A. baumannii*. Further, the degradation mechanism of Antitoxin in the presence and absence of adenosine triphosphate (ATP) molecules are explained through MD simulation. The result reveals that the cleavage of Antitoxin could be possible with the presence of ATP by displaying its response from 20 ns, whereas the Zeta Toxin/Epsilon was unstable after 90 ns. The obtained results demonstrate that Zeta Toxin is "temporarily favorable" for ATP to undergo phosphorylation at UNAG kinase through the substrate tunneling process. The study further evidenced that phosphorylated UNAG prevents the binding of MurA, the enzyme that catalyzes the initial step of bacterial peptidoglycan biosynthesis. Therefore, the present study explores the binding mechanism of Zeta Toxin/Epsilon Antitoxin, which could be beneficial for preventing cell wall biosynthesis as well as for unveiling the alternative treatment options to antibiotics.

 © 2021 Wiley Periodicals LLC

KEYWORDS

Acinetobacter baumannii, molecular docking, molecular dynamics, post segregational killing, UNAG kinase, UNAG-3P, Zeta Toxin-Epsilon Antitoxin

1 | INTRODUCTION

Acinetobacter sps. are Gram-negative, rod-shaped, nonmotile, aerobic bacterium widely distributed in soil and water as free-living saprophytes.[1–4] So far, 63 species of *Acinetobacter* have been identified, among which *Acinetobacter baumannii* was considered as one of the clinically important pathogens and has been emerged as a leading causative agent of nosocomial infections in humans.[5–7] *A. baumannii* harbor multiple mechanisms to survive under adverse environmental conditions, such as soil and water, and are commonly found in hospitals and community settings. Ventilator-associated pneumonia, catheter-associated urinary tract, and bloodstream infections are the most prevalent infections caused by *A. baumannii* with considerable attributable mortality.[3,8] This bacterium mainly colonizes in the skin and is often isolated from the respiratory tract and oropharynx secretion of the infected individuals.[9,10] Infectious Disease Society of America has declared *A. baumannii* as an ESKAPE pathogen that poses a great threat to human health.[2,11] Existing antibiotics are ineffective against multidrug-resistant, pan-resistant, and extensive drug-resistant *A. baumannii* strains, and toward which potential antibiotics are urgently needed.

Toxin–Antitoxin (TA) system was first identified in *Escherichia coli* plasmid in 1982.[12] The TA systems are small genetic modules that are widely distributed in plasmids and chromosomes of prokaryotic bacteria, archaea, and also in unicellular fungi. The bacterial cell stably maintains the copy number of plasmids through several mechanisms and one of which is Post Segregational Killing (PSK).[13–15] The PSK is mediated through the TA system when a plasmid-free daughter cell arises upon cell division; the Antitoxin quickly gets degraded, causing Toxin expression and cell death.[12,16] These TA systems are found in both Gram-positive and Gram-negative bacteria (e.g., relB/E, HicA/B, HipA/B, Zeta, and mazEF) and are widespread in different organisms.[3] Over ~15 TA systems have been reported in *A. baumannii*, which are mostly encoded by plasmid and few of them are encoded by chromosomes.[3,11] Putting all together, the Toxin plays a crucial role, including inhibition of translation through RNA degradation, cell wall biosynthesis, etc.[11] Zeta Toxin-Epsilon Antitoxin was initially identified in *Streptococcus pneumonia* and later in *Streptococcus pyogenes*, *Staphylococcus aureus*, *Enterococcus faecalis*, *Clostridium perfringens*, *Neisseria gonorrhoeae*, *Acinetobacter johnsonii*, etc.[17–20] Based on the interaction profile, the TA systems were generally classified into eight different types.[21,22] The Toxin exists in the form of protein, whereas the Antitoxin may be either in the form of RNA or protein. Antitoxin types I, III, and VIII are small noncoding RNA molecules that regulate the level of Toxin protein either by inhibiting Toxin translation (type I) or through direct inhibition of Toxin protein (type III). While type II, IV, V, VI, and VII Antitoxins are proteins that form a complex with its cognate Toxin, acts as a Toxin antagonist and promotes Toxin degradation.[22] Hence, the present study focused on the type II TA system. A bi-cistronic operon under the control of omega operator codes for Zeta Toxin and Epsilon Antitoxin transcribes both Epsilon and Zeta protein in the form of the inactive complex as shown in Figure 1A.[4,14,19,23,24]

The PSK signal that degrades the Antitoxin (Epsilon) using the ATP-dependent protease, which enables the activity of Zeta Toxin is depicted in Figure 1B.[22,25] The counter-balanced effect of Zeta Toxin was carried out by Epsilon Antitoxin to prevent the access of phosphoryl transferase activity.[12] The Zeta Toxin protein contains a highly conserved Walker-A motif for ATP binding. Upon activation, Zeta Toxin phosphorylates the ubiquitous peptidoglycan precursor uridine diphosphate-*N*-acetylglucosamine (UNAG) with the aid of UNAG kinase and forms UDP-*N*-acetylglucosamine-3′-phosphate (UNAG-3P). The phosphorylation reaction is catalyzed by UNAG kinase in the presence of ATP as co-factor.[2,15,18–21] The UNAG kinase facilitates the transfer of the phosphate group from ATP to the -OH group (C3′) of UNAG, which prevents the peptidoglycan biosynthesis (Figure 1C). Hence, UNAG cannot be utilized by the MurA enzyme for the conversion of UNAG to UNAG-enol pyruvate to proceed with peptidoglycan biosynthesis (Figure 1D) The improper peptidoglycan gets lysed due to the hydrostatic pressure.[18–20,25–27]

The Zeta Toxin of *A. baumannii* shares high homology with Gram-positive (*S. pyogenes* and *S. pneumoniae*) and Gram-negative (*N. gonorrhoeae*) pathogens.[18] Recently, Srivastava et al.,[28] have reported that the unicellular eukaryote *Leishmania donovani* possesses orthologs Zeta-Toxin (Ld_ζ1) which is structurally and functionally similar to the bacterial-Zeta-Toxin (BzT).

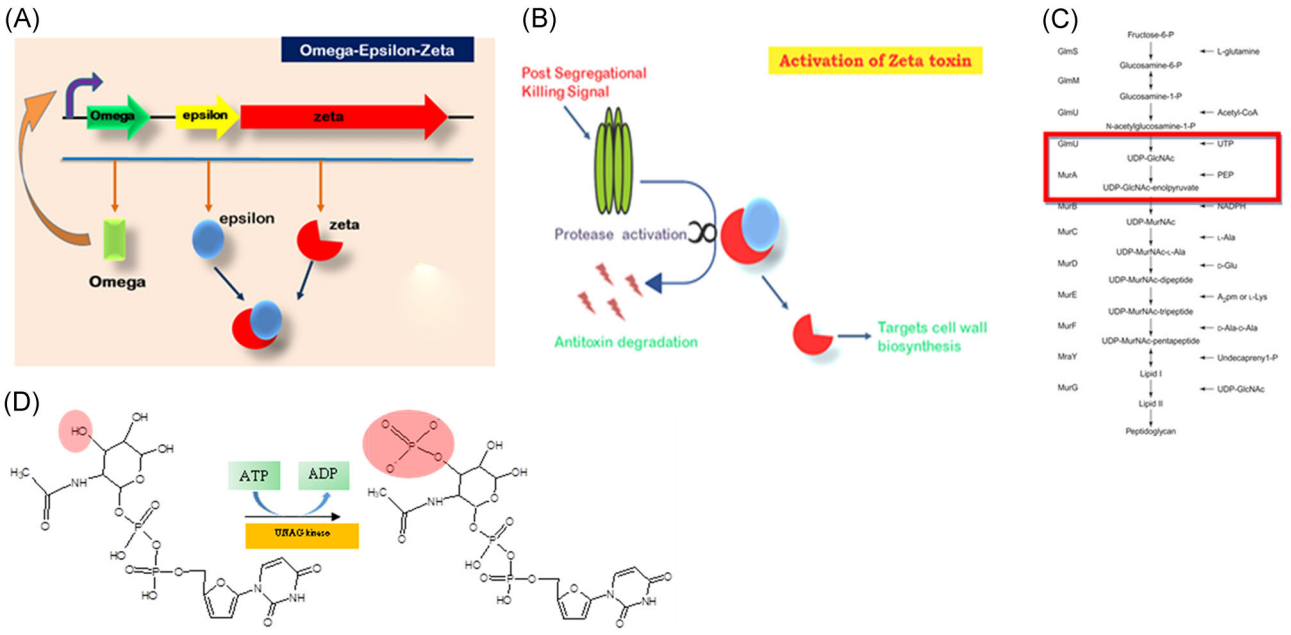


FIGURE 1 (A) Expression of Toxin and Antitoxin in the bacterial system. (B) Schematic representation of Activation of Zeta Toxin. (C) Peptidoglycan Biosynthesis Pathway. (D) Interconversion of UNAG and UNAG-3P. UNAG, uridine diphosphate-*N*-acetylglucosamine; UNAG-3P, UDP-*N*-acetylglucosamine-3′-phosphate

L. donovani harbors similar ATP and UNAG binding pockets like prokaryotic bacteria and triggers cell death.[28] Therefore, Zeta-Toxin mediated cell death can act as a novel anti-eukaryotic and anti-prokaryotic chemotherapeutics. An understanding of the Zeta Toxin and Antitoxin (TA) system is a prerequisite to get better knowledge about the function of the proteinaceous TA system. Therefore, investigating the TA system could be a promising strategy for combating infections mediated by multi-drug resistant *A. baumannii*. The objective of the present study is (i) to identify the presence or absence of Antitoxin in *A. baumannii*, (ii) to evaluate the mode of elimination of Antitoxin either naturally or influenced by ATP, and (iii) to investigate the stability of UNAG-3P in the complex.

2 | MATERIALS AND METHODS

2.1 | Protein modeling

The Zeta Toxin sequence was obtained from the UniProt database using the accession number U5QBG2.[29] The Pfam webserver was used to identify protein domain and families[30] and ProtParam was used to predict the physicochemical properties of the sequence.[31] Initially, BLASTp search was performed against Protein Data Bank (PDB) to select the appropriate template for homology modeling of the Zeta Toxin.[32] Due to the unavailability of a proper template in the PDB, two different ab initio methods, such as RAPTOR[33] and ROBETTA,[34] were employed to predict the protein model. The predicted models are ranked based on the scores assigned by the scoring functions implemented in each tool. Further, the predicted model was examined through the SAVES server to determine the quality of the protein.[35]

2.2 | Molecular dynamics (MD) simulation

The MD simulation was performed for *N. gonorrhoeae* and *S. pyogenes* TA system for both *apo* and complex. Also, MD simulation was performed for the following: (i) Zeta Toxin (*apo*), (ii) Zeta Toxin-Antitoxin (Epsilon), (iii) Zeta Toxin-Antitoxin-ATP, (iv) UNAG kinase (*apo*), and (V) UNAG kinase-UNAG-3P complexes of *A. baumannii* with GROMACS (GROningen MAchine for Chemical simulations) using GROMOS96(43a1) force field.[36] The *apo* and complex forms were solvated on the simple point charge (SPC) in a cubic cell dimension of 1.0 nm.[37] The topology files for the ligand molecules were generated using the PRODRG server.[38] An appropriate balancing counter ion (Na^+ or Cl^-) was added to neutralize the systems. To avoid the steric clashes, the solvated and neutralized system is subject to relaxation through Steepest Decent (SD) minimization algorithm with a

maximum of 50 000 steps.[39] Then, the system was equilibrated with 100 ps of isothermal-isochronic (NVT) at 300 K followed by equilibration using isothermal-isochronic (NPT) ensemble at a pressure of 1 atm for 100 ps. Linear Constraint Solver (LINCS) algorithm was used to converge the bond angles and length in the system.[40,41] Finally, the production MD run of 100 ns was performed for *apo* proteins and 50 ns for the complex (TA) systems. The trajectories obtained from the MD simulations were analyzed to explore the structural parameters, including root mean square deviation (RMSD), root mean square fluctuation (RMSF), and radius of gyration (Rg).[37] All the plots were generated using the XMGRACE program.

2.3 | Detection of Antitoxin

To confirm the presence of Antitoxin, the plasmid sequence online search and TASmania Finder was employed to predict the Antitoxins in *A. baumannii*.[42]

2.4 | Active site prediction and protein-protein docking

The online tools like Residue Depth,[43] CASTP,[44] 3D Ligand site,[45] RAPTOR[33] were used to predict the binding sites of the proteins. Further, protein-protein docking was performed using HADDOCK (High Ambiguity Driven protein-protein Docking)[46] web server. The best-scored model from each cluster is extracted based on the scoring functions embedded in the HADDOCK server.

2.5 | Substrate binding site

Earlier reports state that ATP occupies the conserved P-loop region (which contains the SGK motif) in the Zeta Toxin.[19,47] Further, to identify the presence of the P-loop region in the sequence, the CDART tool was employed.[48] Multiple sequence alignment was performed with other organisms, such as *N. gonorrhoeae* and *S. pyogenes* using ClustalO.[49]

2.6 | UNAG kinase modeling

Maria Moreno-del Alamo et al.[50] have reported that the Zeta Toxin favors provisional binding with the ATP molecule. Moreover, the ATP molecule may be flipped to the UNAG kinase for the conversion of UNAG-3P upon ATP hydrolysis.[18] For structure determination, the UNAG kinase sequence was obtained from UniProt using the accession number M4UT87.[29] The BLASTp[32] search was performed to identify the appropriate template to construct the three-dimensional structure for UNAG kinase. Based on the BLAST search, no structural evidence was obtained. Hence, the structure was predicted using ab initio servers, such as RAPTOR[33] and ROBETTA.[34]

2.7 | Molecular docking

The molecular docking was performed to identify the binding affinity between UNAG and ATP-UNAG kinase complex using Autodock *ver.* 4.2.[51] Before docking, the protein was prepared by adding polar hydrogen and assigning atomic charges. The ligand was prepared by adding Gasteiger charges, non-polar hydrogen atoms, and assigning rotatable bonds. The grid box was generated for the determined active site. The molecular docking was performed using the Lamarckian genetic algorithm (LGA) with the default parameters.[51,52] The results were quantified in the terms of binding affinity. All the figures were generated by using UCSF Chimera.[53] The models showing the highest binding affinity were chosen to examine the conversion of UNAG to UNAG-3P.

3 | RESULT AND DISCUSSION

3.1 | Homology modeling of Zeta Toxin

Zeta Toxin of *A. baumannii* contains 360 amino acid residues and the physicochemical analysis revealed that the protein is stable with the theoretical molecular weight of 41.15 kDa (Figure S1). Moreover, domain analysis results revealed that the amino acid residues starting from 37 to 142 position corresponds to the Zeta Toxin domain family (Figure S2). The BLAST search identified the crystal structure of epsilon_1/Zeta_1 antitoxin/toxin system from *N. gonorrhoeae* in complex with UNAM-4P (PDB: 6EPI) as an appropriate template for our query sequence. Initially, template 6EPI was used to construct the model (query) using the SWISS modeler. The conformational constancy of the generated model is not reasonably good, and this may be due to the huge gaps observed in the sequence alignment in the regions between 1–12 and 233–360. Hence, to obtain the full-length structure with better conformational folding, we have employed two different ab-initio modelers (RAPTOR and ROBETTA) to predict the protein structure. Comparison of the results obtained from two different tools showed that the model generated by ROBETTA is more reliable with the Z-score of 161.885. The predicted three-dimensional structure of Zeta Toxin is shown in Figure 2A, and the Ramachandran

plot computed in the SAVES server indicates that 94.3% of the residues have occupied the most favorable regions while 5.4% appeared in additional allowed regions and none of the amino acid residues occupied the disallowed region (Figure S3). The obtained results reflect the reliability of the predicted model.

3.2 | MD simulation of Zeta Toxin (apo protein)

To understand the conformational flexibility and stability of the predicted structure, MDs were performed for a time period of 100 ns. The RMSD results revealed that initial deviations were observed up to 10 ns followed by a gradual decrease up to 30 ns and then the system reached a stable RMSD value of 0.63 nm in the remaining simulation time (Figure 2B). To explore the residual fluctuations of the amino acid residues during the simulation time, an RMSF plot was generated. Based on the results obtained, the residual fluctuations were observed in the range from 0.1 to 0.6 nm with an average fluctuation around 0.4 nm (Figure 2C). Amino acids, namely Ser16, Glu43, Gln64, Ser131, Glu137, Lys170, Val181, Lys183, Arg190, Asp211, Arg242, Asp248, Glu286, Lys300, Lys303, and Arg320 showed fluctuations during the simulation time. In addition, amino acid residues between the regions (44–50) showed no significant fluctuations. The RMSD of the predicted structure and the final frame of MDs have shown 0.3 Å.

3.3 | Identification of Antitoxin

Zeta Toxin systems are widely distributed in the chromosomes or in the plasmids of various pathogenic bacteria. In the case of *A. baumannii*, the Zeta Toxins are found in the plasmids.[3,12,14] The sequence analysis using tBLASTn showed 100% identity with plasmid pACICU2, which evidenced the presence of the Toxin coding element in the protein. Additionally, the following plasmids, such as pAC30C, pACICU2, p1AB5075, pA105-1, pAb-G7-2, pABTJ1, pAC29b, pD72-2, pMC23.1, and pNDM-AB were retrieved from the literature search;[12,14] and their corresponding accession ID's are given in Table 1. An outcome of TASmania Finder displays the pattern like representation "xxTxxA" under the operon structure, which indicates the presence of Toxin, Antitoxin, and uncharacterized TAS families (Figure S4). Both Toxin and Antitoxin patterns are observed in the following plasmids, including pABTJ1, pACICU2, pD72-2, p1AB5075, pA105-1, and not observed in the remaining plasmids. To identify the variations occurring among the plasmids, the BLASTn was performed. As a result, the pACICU2 has shown 100% sequence similarity with pD72-2 and pABC7-2, whereas the pD72-2 has shown

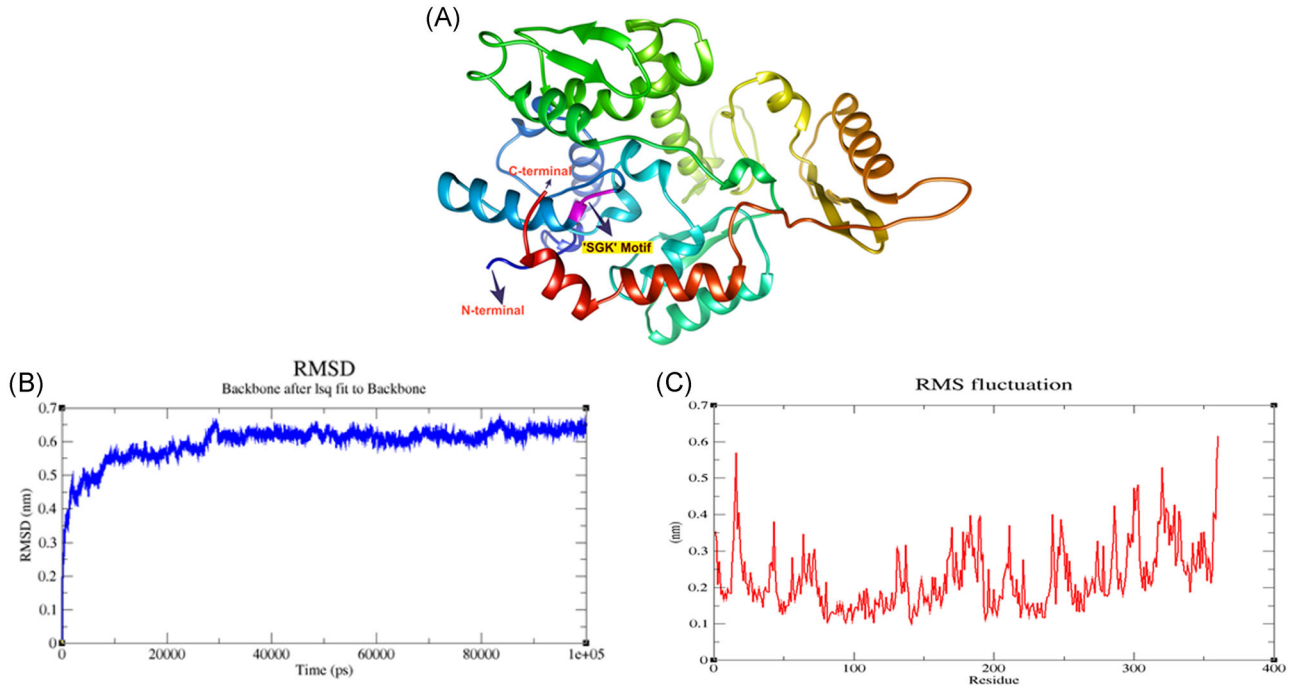


FIGURE 2 (A) 3D Structure of Zeta Toxin modeled using ROBETTA ab initio server. (B) RMSD of Molecular Dynamics Simulations of Zeta Toxin. (C) RMSF of Zeta Toxin. RMSD, root mean square deviation; RMSF, root mean square fluctuation

TABLE 1 List of plasmid identified with Zeta Toxin

Sl. no.	Plasmids	Length of base pairs	Accession ID
1	pAC30C	71 433	CP007580.1
2	pACICU2	70 101	NZ_CP031382.1
3	p1AB5075	83 610	NZ_CP008707.1
4	pA105-1	70 098	NZ_KR535992.1
5	pAb-G7-2	70 100	NC_025104.1:1474-2556
6	pABTJ1	77 528	NC_017848.1
7	pAC29b	74 749	NC_CP008851.1
8	pD72-2	70 102	KM051846.1
9	pMC23.1	67 441	NZ_MK531538.1
10	pNDM-AB	47 098	NC_020818.1

TABLE 2 Similarity among the plasmids using Blastn

Plasmid	Plasmid	Sequence similarity
pACICU2	pAC30C	98.85%
pACICU2	pAC29b	98.86%
pACICU2	pABTJ1	74.86%
pACICU2	pAB85-3	99.35%
pACICU2	pNDMA B	No match
pACICU2	pD72-2	100%
pACICU2	pABC7-2	100%
pD72-2	pAC30C	98.80%
pD72-2	pAC29B	98.81%
pD72-2	p1AB5075	100%
pD72-2	pA105-1	100%
pA105-1	p1AB5075	100%
pABTJ1	pAC30C	74.82%
pABTJ1	pAC29B	99.80%
pAC29b	pAC30C	74.82%
pAC29b	pABG72	99%
pABG72	pAC30C	99.5%

100% similarity with p1AB5075, pA105-1. However, pA105-1 has shown 100% similarity with p1AB5075. The remaining plasmids displayed about 99% similarity while pABTJ1 showed 74% similarity (Table 2). Even though all the sequence possesses 100% sequence similarity between them, the Antitoxin patterns in the sequences cannot be detected.

3.4 | Prediction of the active site and Toxin-Antitoxin docking

Toxin interacts with Antitoxin to perform the essential cellular processes in most of the important bacterial pathogens. Therefore, understanding the binding mechanism of TA association is essential in the aspect of drug discovery. Earlier studies have evidenced the presence of Antitoxin in Gram-negative pathogen (*N. gonorrhoeae*).[17,18,20] The sequence alignment between *N. gonorrhoeae* and *A. baumannii* has shown 36.84% similarity. Based on the obtained result, the crystal structure of the Epsilon/Zeta/Toxin system of *N. gonorrhoeae* (PDB ID: 6EPI) was extracted from the PDB database.[54] The structure contains eight chains (four Toxins and four Antitoxins). The A, C, E, G chains correspond to Antitoxin and B, D, F, H chains represent the Toxin. To examine the interaction between the Zeta Toxin and Epsilon, the TA complexes were separated into four different pairs (AB, CD, EF, and GH). Initially, the AB pair was superimposed with the Zeta Toxin and subsequently, the CD, EF, and GH pairs were superimposed. The H chain of *N. gonorrhoeae* was found similar to the Zeta Toxin of *A. baumannii* and it was surrounded by the G chain of Antitoxin, which contains about 61 amino acids. Therefore, the "G" chain was extracted and further the protein-protein docking was performed against the Zeta Toxin of *A. baumannii*. The active site residues of Zeta Toxin were predicted using residue depth, CastP, and Raptor servers. Based on the predictions, the amino acid residues, such as Thr53, His54, Asp55, Ser56 Gly57, Lys58, Asp59, Val60, Ser81, Lys84, His97, Ile113, Ser114, Asn115, Glu11, Lys117, Tyr118, Val141, Leu151, Lys180, and Val181, were considered as active residues for docking (Table 3). The protein-protein docking was executed using the HADDOCK webserver with the aforementioned active site residues of the Zeta Toxin and whole amino acid residues are selected in the "G" chain. The outcome of the models was sorted based on the HADDOCK scoring function (Z-score). Among all the predicted models, cluster-3 has produced a reliable Z-score of −1.6 kcal/mol (Table 4). Further, the predicted model was superimposed with the G and H chains of *N. gonorrhoeae* resulted in the RMSD value of 0.5 Å. In the predicted TA complex, the amino acid residues of Arg10 and Arg13 in Antitoxin play the important role in the formation of hydrogen bonding interaction with Ser56 and Lys58 of Zeta Toxin (Figure 3A,B). The analysis depicts that the Zeta Toxin of *A. baumannii* may contain the Epsilon Antitoxin similar to its existence in *S. pyogenes* and *N. gonorrhoeae*. The residual interactions between the Zeta Toxin-Antitoxin are shown in Table 5.

TABLE 3 Active site prediction using an online server

Zeta toxin active site prediction servers				
Sl. no.	**Residue depth**	**CAST P**	**3D Ligand site**	**Raptor**
1	Thr53	Thr53	Thr53	
2	His54	His54	His54	His54
3	Asp55	Asp55	Asp55	Asp55
4	Ser56	Ser56	Ser56	Ser56
5	Gly57		Gly57	Gly57
6	Lys58		Lys58	Lys58
7	Asp59		Asp59	Asp59
8	Val60		Val60	Val60
9	Ser81	Ser81		Ser81
10	Lys84	Lys84		Lys84
11	His97	His97		His97
12	Asn113	Asn113		Asn113
13	His114	His114		His114
14	Glu115	Glu115		Glu115
15	Phe116	Phe116		Phe116
16	Lys117	Lys117		Lys117
17	Asn118	Asn118		Asn118
18	Val121	Val121		Val121
19	Val141	Val141	Val141	
20	Leu151	Leu151	Leu151	Leu151
21	Arg180		Arg180	Arg180
22	Val181		Val181	Val181

3.5 | ATP binding regions in the Zeta Toxin

vEarlier studies reported that the *Streptococcal* system forms the heterotetramer (Epsilon-2 Zeta-2), where the Epsilon Antitoxin prevents the binding of ATP in the active site of Zeta Toxin.[17] The existing reports indicate the presence of the "SGK" motif on the surface closer to the active site and it might play an important role in the substrate binding.[17,18,21] Moreover, *N. gonorrhoeae* and *S. pyogenes* have shown the ATP molecule located in the P-loop region by recognizing the specific pattern of the GxxxSGK [TS] motif. The CDART analysis showed the presence of the P-loop region in the Zeta Toxin (Figure S5A). Furthermore, the multiple sequence alignment analysis confirms the conserved sequence pattern of SGK motif in (corresponding 56–58th position) *A. baumannii*, *N. gonorrhoeae*, and *S. pyogenes* (Figure S5B). Molecular docking was also performed to accommodate ATP in the Zeta Toxin (Figures 3C,D). Before the entry of ATP in Zeta Toxin, the two amino acid residues, namely Arg10 and

TABLE 4 Protein–protein docking of Zeta Toxin-Epsilon Anti Toxin using Haddock server

Cluster	Haddock score	RMSD	Van der waals energy	Electrostatic	Desolvation energy	Restraints violation energy	Buried surface area	Z score
3	−68.1 ± 9.8	11.8 ±0.1	−61.9 ± 4.7	−338.1 ± 45.1	1.6 ± 1.9	598.6 ± 95.4	2529.1 ± 175.2	−1.6
10	−61.7 + /− 14.1	3.7 + /− 0.2	−73.0 ± 8.5	−236.3 ± 9.2	−7.5 ± 1.1	662.0 ± 88.5	2806.2 ± 187.1	−1.2
4	−54.8 ± 19.8	3.4 ± 0.2	−62.5 ± 9.0	−294.0 ± 28.9	−0.7 ± 2.8	672.5 ± 124.6	2521.2 ± 225.1	−0.8

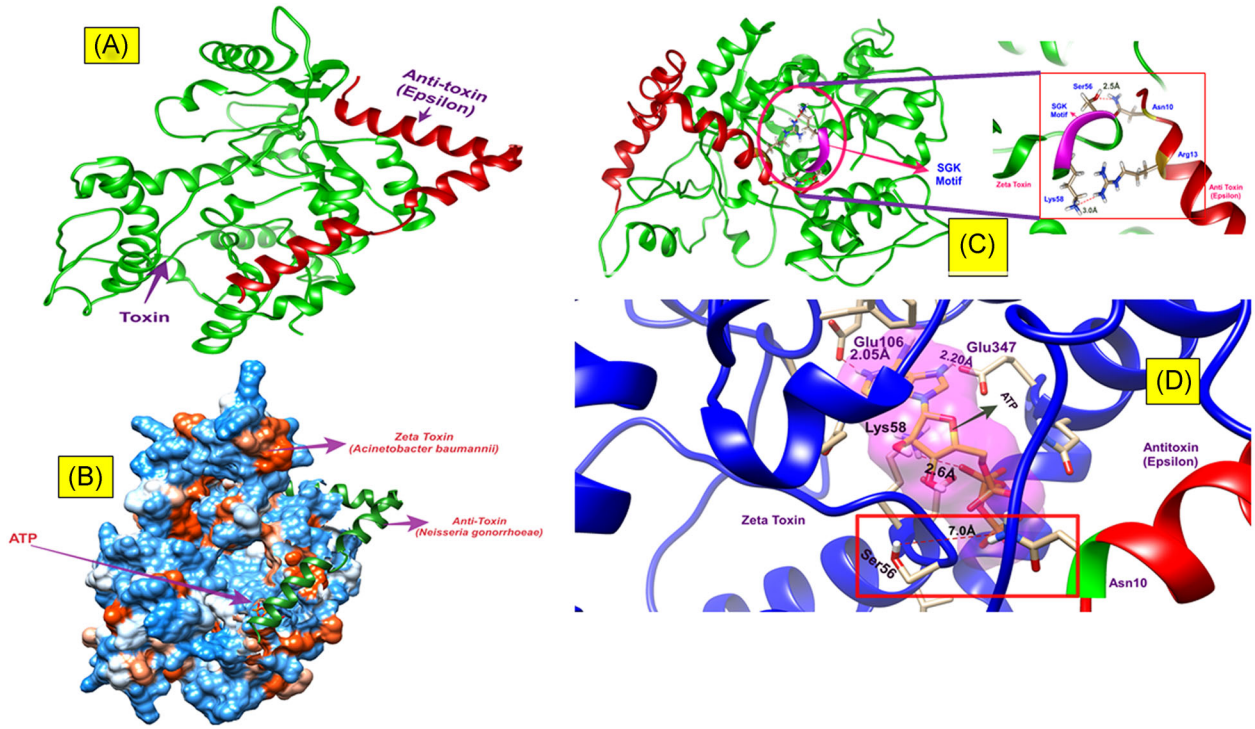


FIGURE 3 Summarizes the entry of ATP in the Zeta Toxin – Epsilon Antitoxin complex. (A) The binding affinity of Antitoxin (Epsilon) with Zeta Toxin in Ribbon model and (B) space fill model representing the presence of ATP binding location. (C) ATP creates an entry in the Zeta Toxin – Epsilon complex. (D) Presence of ATP and showing interactions with Ser56, Lys58, Glu106, and Glu347

TABLE 5 Interactions of Zeta Toxin-Antitoxin (TA) systems

Zeta Toxin	Antitoxin	Distance (Å)
Asn154 CO	Ser9 NH	2.70
Asn154 CO	Asn10 CO	1.86
His54 CO	Asn10 NH	1.91
Asp55 CO	Arg13 NH	1.60
Ser56 CO	Asn10 NH	2.50
Lys58 CO	Arg13NH	3.00
Glu338 CO	Lys21 NH	1.96
Lys117 CO	Ser44 NH	2.31
Asn118 NH	Arg38 CO	2.20
Asp341 CO	Arg24 NH	1.62
Asp34 CO	Arg24 NH	2.38
Lys337 NH	Glu34 CO	1.70
Asp124 CO	Arg38 NH	1.69

Asn13 of Antitoxin holds the interaction with Ser56 and Lys58 of the "SGK motif" of Zeta Toxin maintaining the distance of 2.5 and 3.0 Å, respectively. Rocker et al.[19] reported that the binding site of Zeta Toxin was blocked by the Epsilon Antitoxin. The present study explores that the ATP could be accommodated within the Zeta Toxin when it is complexed with Epsilon Antitoxin. The same contacts, such as Ser56-Asn10 and Lys58-Arg13, were extended up to 7.0 Å and these gaps facilitate the ATP entry into the "P"-loop region. Moreover, other residues, such as Glu55, Glu106, and Glu347, have shown interactions with the ATP residues and the Lys58 showed significant interaction with the "O" atom attached in the P2 of ATP. Hence, the study confirms that the SGK region of Zeta Toxin and Arg10, Asn13 of Antitoxin are considered as crucial residues for an ATP entry. Therefore, further experimental investigations are required to understand the impact of ATP in the presence of Zeta Toxin complex with Epsilon Antitoxin.

3.6 | MD simulation of Toxin-Antitoxin system

3.6.1 | MD simulation of *N. gonorrhoeae and S. pyogenes*

Initially, the MDs were performed for *N. gonorrhoeae and S. pyogenes* TA system (*apo*) for 100 ns and in the presence of ATP for 50 ns. The *S. pyogenes* in *apo* form shows stability at the beginning of the MD simulation and then slight

disturbance was observed between 18 and 41 ns. Further, the system got rigorously disturbed after 50 ns till the end of the simulation, which was observed from RMSD analysis (Figure S7A). Whereas the RMSD of *N. gonorrhoeae apo* form showed series fluctuations after 22 ns and it was observed throughout the simulation (Figure S7C). On the contrary, in the presence of ATP, the Toxin–ATP–Anti Toxin in *S. pyogenes* showed a raise in RMSD peak up to 0.8 nm till 20 ns and the peak maintains at 0.6 nm. Hence, the RMSD plot shows that the simulation is somehow brittle. The RMSD plot of *N. gonorrhoeae* Toxin–ATP–Anti Toxin complex has shown unstable condition at the beginning of simulation by resulting in the peak which raised up to 3.4 nm. The RMSD plot shows that the Anti-Toxin was cleaved around 23 ns and it could be the last where the disturbance of the peak was not continued for the major period of time as it gets firm after 25 ns (Figure S8A). This induces us to investigate further, whether the ATP influences the TA system or it will be eliminated mechanically. Keeping both pathogens as a reference, we have performed a simulation for Zeta Toxin–ATP–Anti Toxin in *A. baumannii*.

3.6.2 | MD simulation of Zeta Toxin-Anti Toxin system

The MD simulation of TA systems was performed for both the complexes, namely (a) Toxin-Antitoxin complex and (b) TA complex with ATP (Toxin-ATP-Antitoxin complex). The RMSD has shown the constructive phase during the simulation as depicted in Figure 4A. The RMSD of the Zeta Toxin and Antitoxin complex sustains between 3 and 4 nm throughout the simulation. The outcome of the simulation analysis indicates that the system was found stable up to 90 ns; later on, it showed large fluctuation till the end of the simulation. Figure 4B,C represents the comparison between the initial frame and the extracted complex, where the separation of the complex was obtained at 90 ns of MDs.

3.6.3 | MD simulation of Zeta Toxin–ATP–Anti-Toxin

As per the hypothesis, we have also performed the MD simulation to understand its significance with the presence of ATP in the TA system (Figure 5A). The Zeta Toxin–ATP–Anti Toxin simulation study reveals that the complex was equilibrated in the early stages of the simulation, then the Antitoxin was expelled from the complex and the ATP was located in the ATP binding pocket of Zeta Toxin till 47 ns. Furthermore, the ATP will be migrated toward the SGK motif of UNAG kinase to undergo phosphorylation. In the close examination, the study revealed that the ATP molecule influences the Antitoxin to separate from the Zeta Toxin (Figure 5A). The uncoupling begins in the 20 ns, where the complex combines at 30 ns. When compared with the 20 and

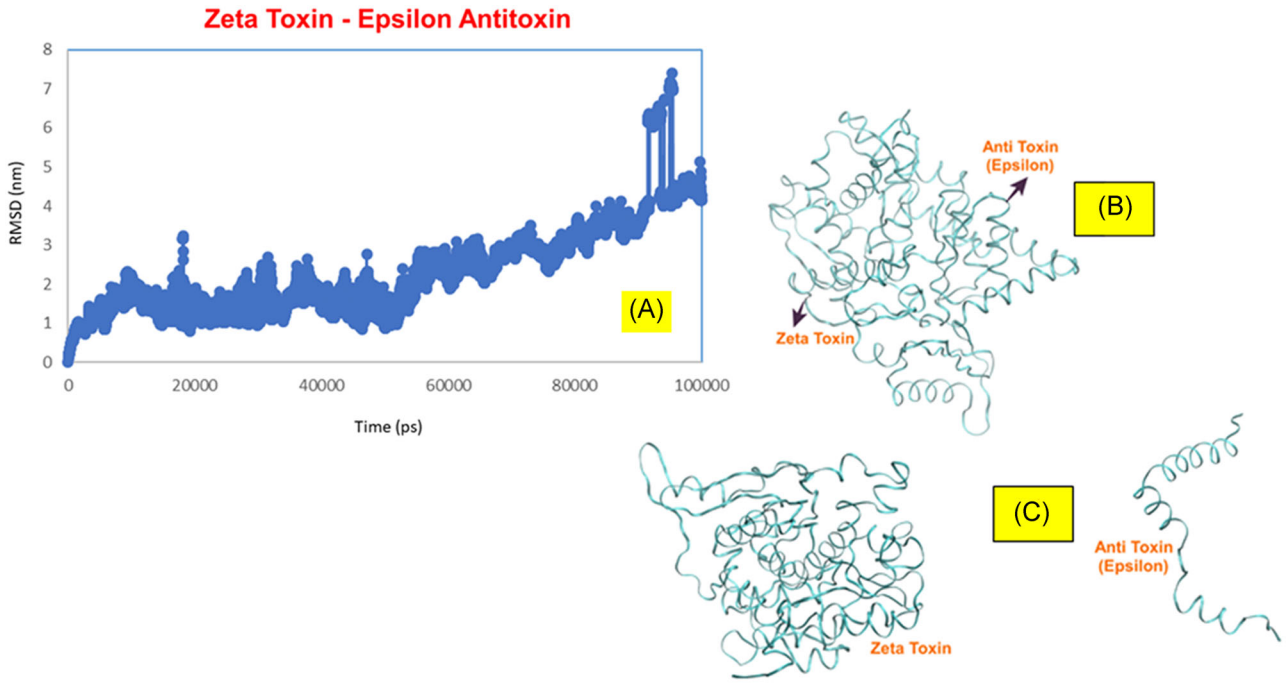


FIGURE 4 Molecular dynamics simulation of Zeta Toxin and Epsilon (A) in the apo-form, (B) Toxin-Antitoxin in the bound form, and (C) cleavage form of Antitoxin from Toxin

30 ns, the ATP molecule is accommodated inside the complex at 20 ns where the ATP molecule pushes the Anti-Toxin in the 33 ns. Then, the Anti Toxin was separated from the complex and it is observed at 40 and 47 ns. Upon the release of Anti Toxin, the ATP also detached from Zeta Toxin, considered as a temporary favorable location. The larger RMSD variation observed in the MD simulation for the complex indicates the instability whereas the small RMSD variation determines the stability of the complex.[55] In both the systems, the larger gap observed in the RMSD analysis indicates the system instability (Figures 4A and 5A). Figure 5B indicates the initial frame and Figure 5iii–iv specifies the extracted complex of 10th, 20th, 30th, 40th, 47th, and 50th ns, respectively, of the MDs, shows some divergence in the RMSD analysis. The overall results suggest that the disturbed hydrogen bonding interactions, as well as nonbonding interactions, influence the separation of Zeta Toxin and Antitoxin complex, which may result in maximum fluctuation. Hence, this analysis witnesses the instability of Epsilon Antitoxin over Zeta Toxin and explores the possibility of Zeta Toxin to adopt a classical phosphotransferase fold (P-loop motif), which is important for the ATP binding.[21]

3.7 | Substrate tunneling

The molecule switches from one enzyme to the catalytic center of the other enzyme without disturbing the environment is otherwise called substrate channeling or substrate tunneling.[56] Figure 6A depicts the separation of ATP and Antitoxin from the Zeta Toxin-Antitoxin-ATP complex, which was observed during the MD simulation. The previous studies reported that the protein contains the canonical Zeta domain, which influences the UNAG kinase activity. Moreover, Maria Moreno-del Alamo et al.[50] have demonstrated that due to the presence of a large cavity, the Zeta Toxin holds the ATP temporarily and resides in the SGK motif of UNAG kinase through substrate channeling (Figure 6B). Therefore, the ATP molecule could be flipped to the UNAG kinase (similar structural characteristics), which leads to the formation of UNAG-3P.

3.8 | Modeling of UNAG kinase

The UNAG kinase protein contains 570 amino acids and the crystallographic structure was not yet solved. The BLASTp search against the PDB has shown only 28% of similarity with *S. pyogenes* and *S. pneumoniae*. Hence, the structure was predicted by using the ab initio servers (ROBETTA and RAPTOR). The results obtained from two different tools showed that the model generated by ROBETTA is more reliable with the *Z*-score of 265.026. The predicted 3D structure was subjected to Ramachandran plot analysis using the SAVES server, which indicates that 90% of the residues are occupied in the allowed regions, and the predicted structure is shown in Figure 7A.

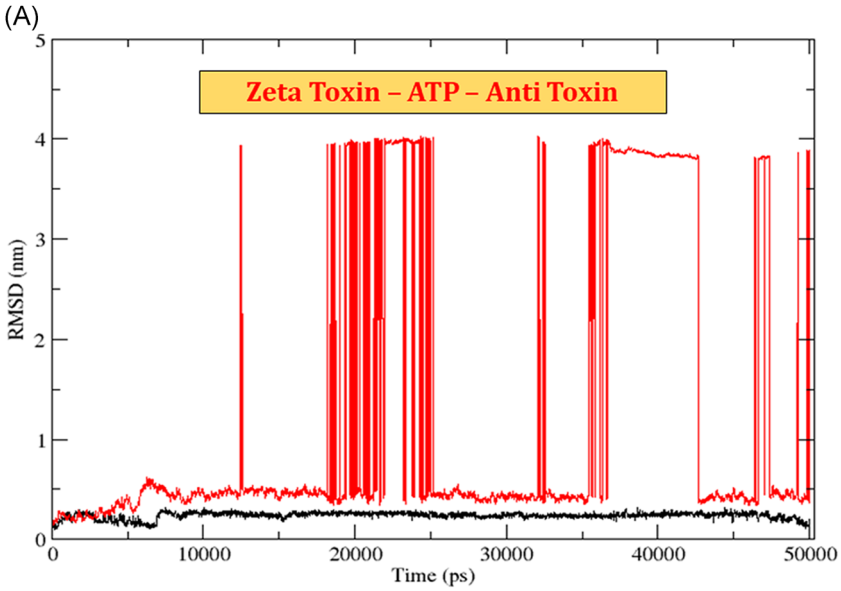


FIGURE 5 (A) RMSD analysis of molecular dynamics simulation of Zeta Toxin and Epsilon in the presence of ATP, (B) (i–vi) structures are obtained at different frames. ATP, adenosine triphosphate; RMSD, root mean square deviation

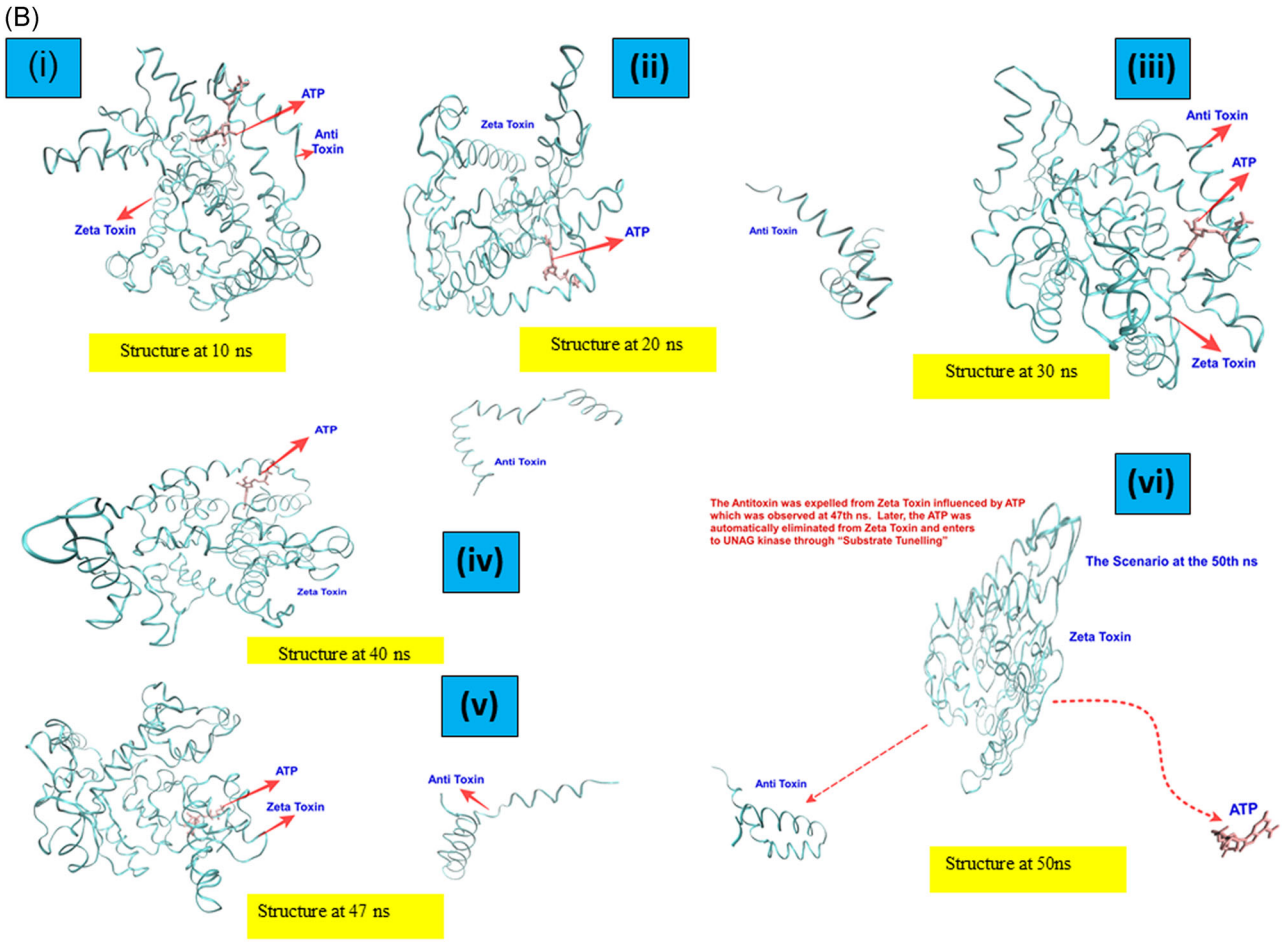


FIGURE 5 Continued

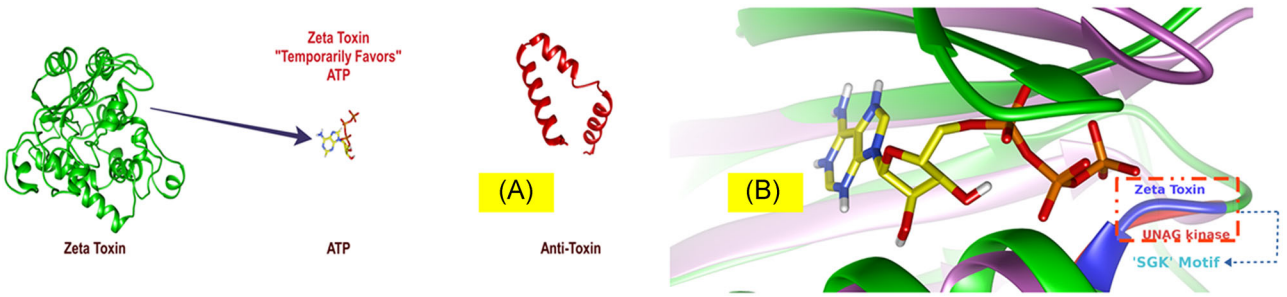


FIGURE 6 (A) Conformation of Zeta Toxin–Antitoxin–ATP complex, which temporarily favors ATP and (B) the substrate tunneling of ATP from Zeta Toxin to UNAG kinase. ATP, adenosine triphosphate; UNAG, uridine diphosphate-*N*-acetylglucosamine

3.8.1 | UNAG kinase MD simulation

To gain insight into the stability and dynamic behavior of the UNAG kinase (*apo* form), it was subjected to 50 ns MD simulation. The RMSD analysis of the backbone has shown deviations during 10–12 ns of the simulation (Figure 7B). The UNAG kinase protein structure has shown slight variations for about 0.2 nm from the baseline during the remaining period of simulation. The superimposed first and final frame of UNAG kinase has shown the RMSD of 0.6 Å. The RMSF plot exhibits the fluctuations for a majority of the residues and raised the peak up to 1 nm (Figure 7C). The

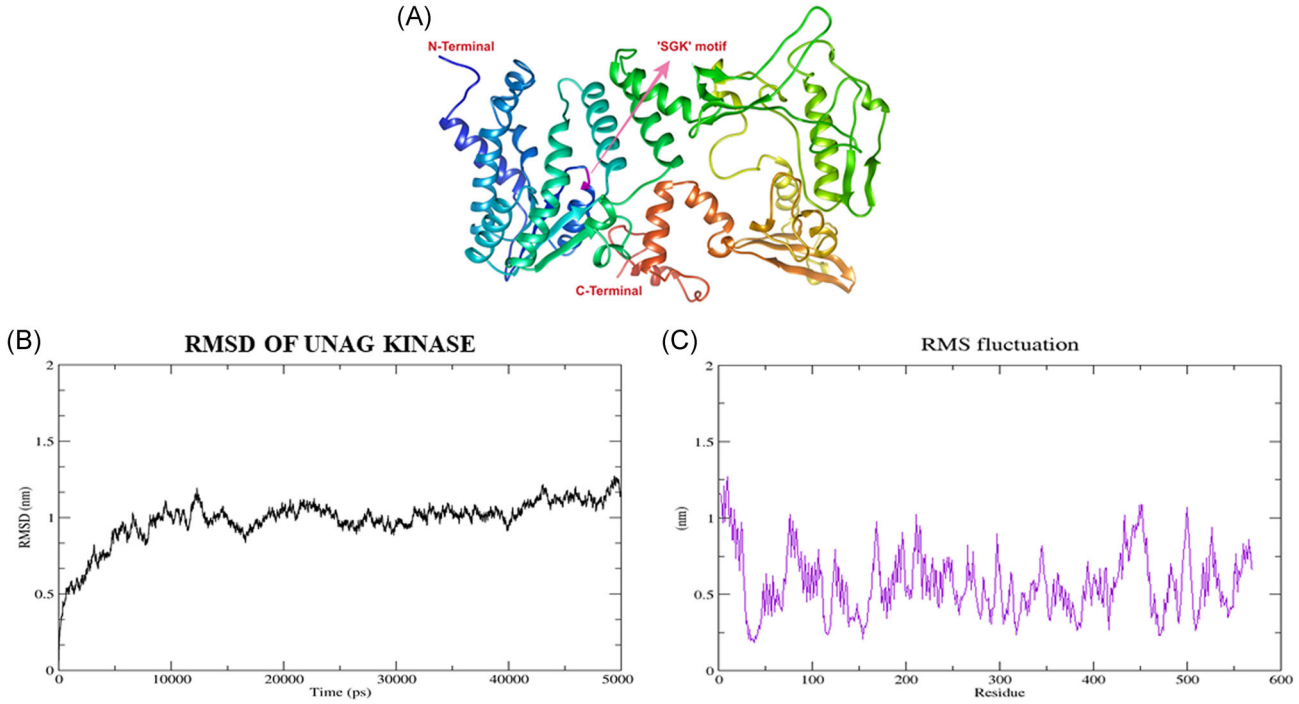


FIGURE 7 (A) UNAG kinase structure modeled using ROBETTA server, (B) Molecular Dynamics simulation of the UNAG kinase RMSD analysis, and (C) RMSF analysis. RMSD, root mean square deviation; UNAG, uridine diphosphate-*N*-acetylglucosamine

RMSD and RMSF do not show much fluctuations during the simulation and this indicates the stability of the protein.

3.8.2 | Formation of UNAG-3P

The Zeta Toxin temporarily favors the binding of ATP and assists the UNAG kinase for the phosphotransfer.[50] The active site residue of UNAG kinase remains the same as the ATP binding site of Zeta Toxin. The active site residues obtained from GxxxSGK[TS] motif pattern were observed through the multiple sequence alignment, which was performed with the sequences of Zeta Toxin, UNAG kinase, and *S. pyogenes* to identify the conserved regions (Figure S6). Further, molecular docking was performed for the UNAG kinase to accommodate both ATP and UNAG. First, the ATP was docked into the "SGK" motif of the UNAG kinase. Mutschler et al.[18] reported that the residues, such as Asp67, Thr120 along with the SGK motif were considered as active site residues for UNAG binding.[18] Hence, the residues, such as Gly39, Gly40, Gly41, Pro42, Ala43, **Ser44, Gly45, Lys46**, Gly47, Leu49, Asp67, and Thr120 are taken as active site residues for UNAG kinase. Importantly, the docked complex of the terminal atoms of ATP and UNAG should face each other with a distance of 3.5 Å, as described by Moreno Del Alamo et al.[50] From the outcome of molecular docking analysis, the better conformation of UNAG in the ATP–UNAG kinase bound complex was obtained with the docking score of −13.5 kcal/mol, and a distance of 3.5 Å was observed between the two terminal atoms (Figure 8A). The obtained distance is practically feasible for the phosphotransfer. The MM/GBSA analysis was performed to calculate the binding free energy between receptor and ligand using the Prime module of Schrödinger LLC, NY, USA.[57] The free energy (ΔG_{bind}) of the UNAG against the UNAG kinase–ATP complex was found to be −36.3379 kcal/mol. Rocker et al.[19] revealed that targeting the UNAG-4P inhibits the peptidoglycan biosynthesis through MurC to MurF. Since the phosphotransfer is a covalent interaction; the terminal phosphate of ATP will be abstracted by UNAG to form a death end product. Therefore, the covalent docking procedure was implemented using CovDock of Schrödinger, LLC, NY[58] to examine whether the phosphate group will be abstracted either by C3′-OH or C4′-OH. As a result of covalent docking, the UNAG-3P favors our study by showing the docking score of −4.8 kcal/mol and the UNAG-4P has shown −2.4 kcal/mol (Figure 8B). Therefore, the formation of the death end product UNAG-3P inhibits MurA that could prevent peptidoglycan biosynthesis.[17–19]

3.8.3 | MD simulation of UNAG-3P complex

To understand the stability of the death end product, the MD simulation was performed for 50 ns. The MD

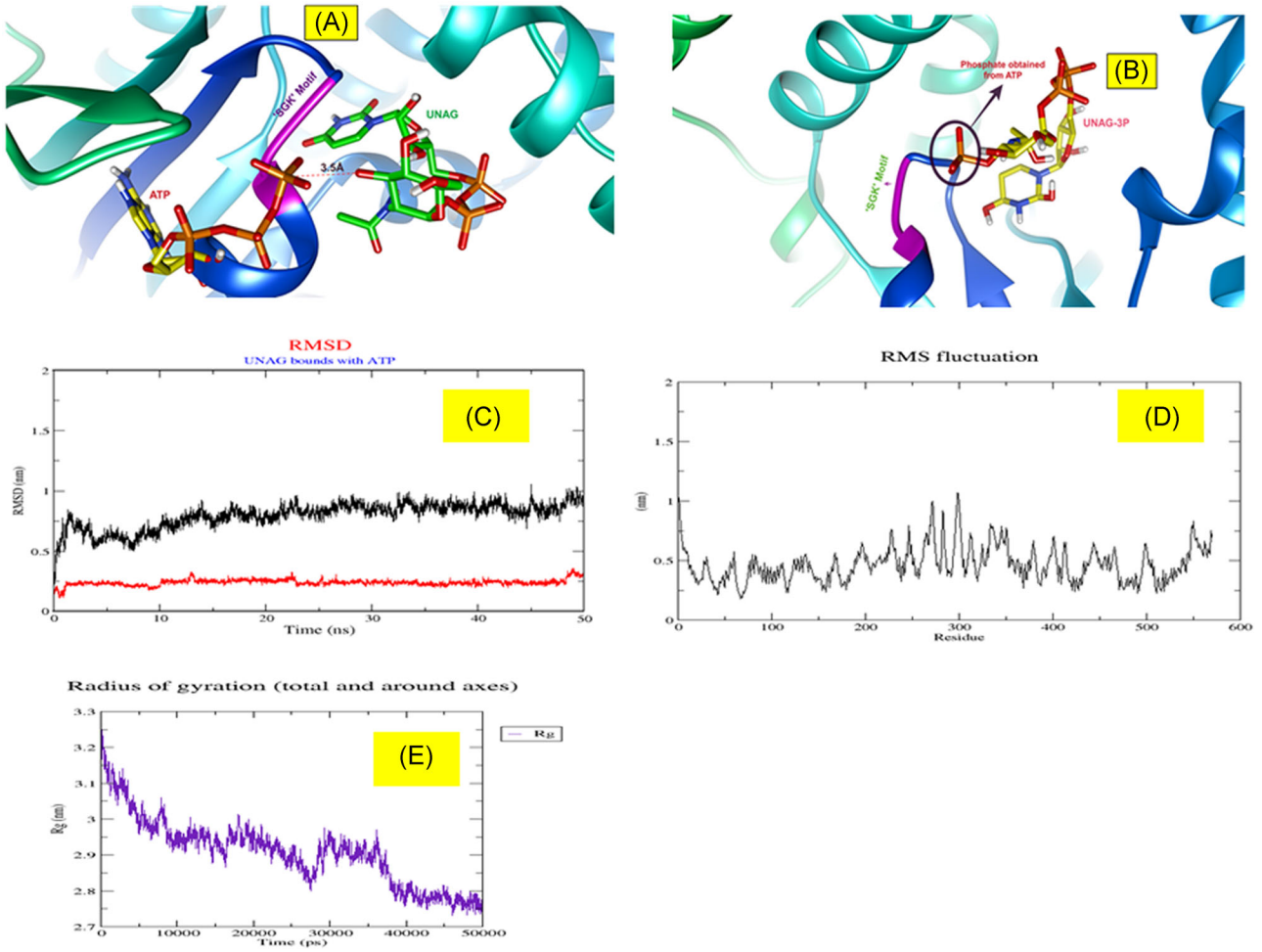


FIGURE 8 (A) The docked complex of UNAG when the ATP bounds with UNAG, showing the distance of 3.5 Å, (B) Formation of UNAG3P in UNAG kinase, (C) RMSD of UNAG kinase (Black curve)–UNAG3P (Red curve), and (D) RMSF of UNAG kinase–UNAG-3P (E) R_g of UNAG kinase–UNAG-3P. ATP, adenosine triphosphate; UNAG, uridine diphosphate-*N*-acetylglucosamine; UNAG-3P, UDP-*N*-acetylglucosamine-3′-phosphate

trajectories of RMSD, RMSF, and Rg were evaluated to ensure the stability of the protein in a regular time-scale in an aqueous environment. The MDs revealed that the Cα of UNAG-3P was stable throughout the simulation. The minor structural variations were observed in the RMSD analysis of UNAG kinase until 10 ns. Subsequently, the RMSD was elevated up to 0.75 nm and stabilized at 0.65 nm throughout the end of the simulation. Whereas, the UNAG-3P in the complex has shown minor fluctuations and maintains the average peak at 0.25 nm (Figure 8C). Hence, the results obtained from the MDs show the stability of the complexes. The RMSF analysis disclosed the information on flexible regions of UNAG kinase-UNAG-3P complexes. The following residues, such as Asp228, Arg248, Lys273, Thr285, Asp300, Arg312, Arg334, His336, Glu346, Asp350, and Arg550, are involved in the fluctuations, ranging between 0.5 and 1 nm (Figure 8D). These fluctuations could be attributed to the presence of the PO^{4-} atom, which has a high prospect to attract the basic amino acid residues, such as Arg, Lys, and His (containing –NH group), and polar amino acids, such as Thr and Asn. On the contrary, the protonated atoms in the ATP are having possible interaction with the carboxylic group of Glu and Asp residues. Therefore, the structural motion of these amino acids may influence the fluctuation and could have interactions with the oxygen and hydroxyl group of UNAG-3P. The R_g is measured to check the compactness of protein and to explore the influence of ligand toward conformational changes that occurred in the protein.[39,40] Moreover, the R_g plot for the complex starts at 3.25 nm and has attained stability at 2.6 nm during the simulation time frame (Figure 8E). The lower value indicates the compactness of the UNAG kinase-UNAG-3P complex. The present study suggests that UNAG-3P is stable and it has higher feasibility for the conversion to UNAG-3P

resulting in the prevention of binding with MurA, which is evident from the MD simulation. Hence, the plausible mechanism is having a good agreement with the earlier reports.[22]

4 | CONCLUSION

The present study evidenced that the TA system has undergone selective evolutionary relationships to remain confined within the Gram-positive as well as Gram-negative bacteria. Taken together, this study reports the identification of Antitoxin. The present study provides more insights to understand Zeta Toxin of *A. baumannii* to have an evolutionary relationship with the Gram-positive and Gram-negative bacterium. The highlights of the study include the following themes, namely (i) understanding the plausible mechanism to identify the free energy of ATP-UNAG complexes, and (ii) identifying specific Antitoxin for the corresponding Zeta Toxin. The MD simulation studies show the instability of the complex Zeta Toxin and Antitoxin, which resulted in the initial stage in the presence of ATP whereas the separation was observed at 90 ns in the absence of ATP. The study also provides novel insights that Zeta Toxin holds ATP temporarily and later it travels toward UNAG kinase to undergo phosphorylation. Therefore, the predicted results illustrate the catalytic roles of Zeta Toxin, which will induce the bacteriostatic activity to prevent the peptidoglycan biosynthesis. These projected outcomes will be a platform to carry out further investigations to address its activity.

ACKNOWLEDGMENTS

The authors BR, SR, and JJ acknowledge the financial grant supported by DST–FIST (SR/FST/LSI-667/2016) (C), DST–Promotion of University Research and Scientific Excellence (PURSE) (No. SR/PURSE Phase2/38) (G), 2017, UGC-RA-2016-18-OB-AM-7124 dated: 18.04.2016, DST-SERB (File No. EMR/2016/000498 dated: 26.09.2016), MHRD-RUSA 2.0, New Delhi (F.24-51/2014-U, Policy (TNMulti-Gen), Dept. of Edn. Govt. of India, Dt. 09.10.2018), DST Indo-Taiwan project order no (GITA/DST/TWN/P-86/2019 dated: 04/03/2020), ICMR Sanction Order Ref. No. (BIC/12(07)/2015 dated: 28.07.2017), and Alagappa University for providing computational facilities to carry out this study.

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

ORCID

Jeyaraman Jeyakanthan http://orcid.org/0000-0002-4594-9610

REFERENCES

1. Chan AP, Sutton G, DePew J, et al. A novel method of consensus pan-chromosome assembly and large-scale comparative analysis reveal the highly flexible pan-genome of *Acinetobacter baumannii*. *Genome Biol*. 2015;16(1):143.
2. Choi CH, Lee JS, Lee YC, Park TI, Lee JC. *Acinetobacter baumannii* invades epithelial cells and outer membrane protein A mediates interactions with epithelial cells. *BMC Microbiol*. 2008;8:216.
3. De Bruijn FJ, ed. *Stress and Environmental Regulation of Gene Expression and Adaptation in Bacteria*. John Wiley & Sons; 2016.
4. Goeders N, Van Melderen L. Toxin-Antitoxin systems as multilevel interaction systems. *Toxins*. 2014;6(1):304-324.
5. Antunes LC, Visca P, Towner KJ. *Acinetobacter baumannii*: evolution of a global pathogen. *Pathog Dis*. 2014;71(3):292-301.
6. Carvalheira A, Gonzales-Siles L, Salvà-Serra F, et al. Acinetobacter portensis sp. nov. and *Acinetobacter guerrae* sp. nov., isolated from raw meat. *Int J Syst Evol Microbiol*. 2020;70(8):4544-4554.
7. Cisneros JM, Reyes MJ, Pachón J, et al. Bacteremia due to *Acinetobacter baumannii*: epidemiology, clinical findings, and prognostic features. *Clin Infect Dis*. 1996;22(6):1026-1032.
8. Fernández-García L, Blasco L, Lopez M, et al. Toxin-Antitoxin systems in clinical pathogens. *Toxins*. 2016;8(7):227.
9. Jurenaite M, Markuckas A, Suziedeliene E. Identification and characterization of type II toxin-Antitoxin systems in the opportunistic pathogen *Acinetobacter baumannii*. *J Bacteriol*. 2013;195(14):3165-3172.
10. Vrancianu CO, Gheorghe I, Czobor IB, Chifiriuc MC. Antibiotic resistance profiles, molecular mechanisms and innovative treatment strategies of *Acinetobacter baumannii*. *Microorganisms*. 2020;8(6):935.
11. Lean SS, Yeo CC, Suhaili Z, Thong KL. Comparative genomics of two ST 195 Carbapenem-resistant *Acinetobacter baumannii* with different susceptibility to polymyxin revealed underlying resistance mechanism. *Front Microbiol*. 2016;6:1445.
12. Ghafourian S, Raftari M, Sadeghifard N, Sekawi Z. Toxin-Antitoxin systems: classification, biological function and application in biotechnology. *Curr Issues Mol Biol*. 2014;16:9-14.
13. Meinhart A, Alonso JC, Sträter N, Saenger W. Crystal structure of the plasmid maintenance system Epsilon/zeta: functional mechanism of toxin zeta and inactivation by Epsilon 2 zeta 2 complex formation. *Proc Natl Acad Sci U S A*. 2003;100(4):1661-1666.
14. Mutschler H, Meinhart A. Type II toxin-antitoxin loci: the epsilon/zeta family. In: Gerdes K, ed. *Prokaryotic Toxin-Antitoxins*. Berlin, Heidelberg: Springer; 2013:205-223.
15. Pandey DP, Gerdes K. Toxin-antitoxin loci are highly abundant in free-living but lost from host-associated prokaryotes. *Nucleic Acids Res*. 2005;33(3):966-976.
16. Tsilibaris V, Maenhaut-Michel G, Mine N, Van Melderen L. What is the benefit to Escherichia coli of having multiple toxin-Antitoxin systems in its genome. *J Bacteriol*. 2007;189(17):6101-6108.
17. Chan WT, Moreno-Córdoba I, Yeo CC, Espinosa M. Toxin-Antitoxin genes of the Gram-positive pathogen *Streptococcus pneumoniae*: so few and yet so many. *Microbiol Mol Biol Rev*. 2012;76(4):773-791.

18. Mutschler H, Gebhardt M, Shoeman RL, Meinhart A. A novel mechanism of programmed cell death in bacteria by toxin-Antitoxin systems corrupts peptidoglycan synthesis. *PLoS Biol*. 2011;9(3):e1001033.
19. Rocker A, Peschke M, Kittilä T, Sakson R, Brieke C, Meinhart A. The ng_ζ1 toxin of the *gonococcal* Epsilon/Zeta Toxin/Antitoxin system drains precursors for cell wall synthesis. *Nat Commun*. 2018;9(1):1686.
20. Schuster CF, Bertram R. Toxin-antitoxin systems of Staphylococcus aureus. *Toxins*. 2016;8(5):140.
21. Amera GM, Khan RJ, Pathak A, Jha RK, Muthukumaran J, Singh AK. Computer aided ligand based screening for identification of promising molecules against enzymes involved in peptidoglycan biosynthetic pathway from *Acinetobacter baumannii*. *Microb Pathog*. 2020;147:104205.
22. Zielenkiewicz U, Kowalewska M, Kaczor C, Ceglowski P. *In vivo* interactions between toxin-Antitoxin proteins Epsilon and zeta of *streptococcal* plasmid pSM19035 in *Saccharomyces cerevisiae*. *J Bacteriol*. 2009;191(11):3677-3684.
23. Lee CR, Lee JH, Park M. Biology of *Acinetobacter baumannii*: pathogenesis, antibiotic resistance mechanisms, and prospective treatment options. *Front Cell Infect Microbiol*. 2017;7:55.
24. Song S, Wood TK. A primary physiological role of toxin/antitoxin systems is phage inhibition. *Front Microbiol*. 2020;11:1895.
25. Brzozowska I, Zielenkiewicz U. The ClpXP protease is responsible for the degradation of the Epsilon antidote to the Zeta Toxin of the *streptococcal* pSM19035 plasmid. *J Biol Chem*. 2014;289(11):7514-7523.
26. Abriouel H, Pérez Montoro B, de la Fuente Ordoñez JJ, Lavilla Lerma L, Knapp CW, Benomar N. New insights into the role of plasmids from probiotic Lactobacillus pentosus MP-10 in Alorena table olive brine fermentation. *Sci Rep*. 2018;9(1):10938.
27. Kumar P, Saumya KU, Giri R. Identification of peptidomimetic compounds as potential inhibitors against MurA enzyme of *Mycobacterium tuberculosis*. *J Biomol Struct Dyn*. 2020;38(17):4997-5013.
28. Srivastava A, Garg S, Jain R. Identification and functional characterization of a bacterial homologue of Zeta toxin in *Leishmania donovani*. *FEBS Lett*. 2019;593(11):1223-1235.
29. UniProt Consortium. UniProt: the universal protein knowledgebase. *Nucleic Acids Res*. 2018;46(5):2699.
30. Finn RD, Bateman A, Clements J. Pfam: the protein families database. *Nucleic Acids Research, 42(Database issue). Nucleic Acids Res*. 2014;42:D222-D230.
31. Daniels LK. Rapid in-office and in-vivo desensitization of an injection phobia utilizing hypnosis. *Am J Clin Hypn*. 1976;18(3):200-203.
32. Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ. Basic local alignment search tool. *J Mol Biol*. 1990;215(3):403-410.
33. Wang S, Li W, Liu S, Xu J. RaptorX-Property: a web server for protein structure property prediction. *Nucleic Acids Res*. 2016;44(W1):W430-W435.
34. Kim DE, Chivian D, Baker D. Protein structure prediction and analysis using the Robetta server. *Nucleic Acids Res*. 2004;32(suppl_2):W526-W531.
35. Laskowski RA, Rullmannn JA, MacArthur MW, Kaptein R, Thornton JM. AQUA and PROCHECK-NMR: programs for checking the quality of protein structures solved by NMR. *J Biomol NMR*. 1996;8(4):477-486.
36. Pronk S, Páll S, Schulz R. GROMACS 4.5: a high-throughput and highly parallel open source molecular simulation toolkit. *Bioinformatics (Oxford, England)*. 2013;29(7):845-854.
37. Rajamanikandan S, Srinivasan P. Pharmacophore modeling and structure-based virtual screening to identify potent inhibitors targeting LuxP of *Vibrio harveyi*. *J Recept Signal Transduct*. 2016;36(6):617-632.
38. Van Aalten DM, Bywater R, Findlay JB, Hendlich M, Hooft RW, Vriend G. PRODRG, a program for generating molecular topologies and unique molecular descriptors from coordinates of small molecules. *J Comput Aided Mol Des*. 1996;10(3):255-262.
39. Amala M, Rajamanikandan S, Prabhu D, Surekha K, Jeykanthan J. Identification of anti-filarial leads against aspartate semialdehyde dehydrogenase of Wolbachia endosymbiont of *Brugia malayi*. *J Biomol Struct Dyn*. 2019;37(2):394-410.
40. Hess B. P-LINCS: a parallel linear constraint solver for molecular simulation. *J Chem Theory Comput*. 2008;4(1):116-122.
41. Yoshii N, Andoh Y, Okazaki S. Fast multipole method for three-dimensional systems with periodic boundary condition in two directions. *J Comput Chem*. 2020;41(9):940-948.
42. Akarsu H, Bordes P, Mansour M, Bigot DJ, Genevaux P, Falquet L. TASmania: a bacterial Toxin-Antitoxin systems database. *PLoS Comput Biol*. 2019;15(4):e1006946.
43. Tan KP, Varadarajan R, Madhusudhan MS. DEPTH: a web server to compute depth and predict small-molecule binding cavities in proteins. *Nucleic Acids Res*. 2011;39:W242-W248.
44. Binkowski TA, Naghibzadeh S, Liang J. CASTp: computed atlas of surface topography of proteins. *Nucleic Acids Res*. 2003;31(13):3352-3355.
45. Wass MN, Kelley LA, Sternberg MJ. 3DLigandSite: predicting ligand-binding sites using similar structures. *Nucleic Acids Research, 38(Web Server issue)*. 2010;38:W469-W473.
46. Dominguez C, Boelens R, Bonvin AM. HADDOCK: a protein-protein docking approach based on biochemical or biophysical information. *J Am Chem Soc*. 2003;125(7):1731-1737.
47. Saraste M, Sibbald PR, Wittinghofer A. The P-loop—a common motif in ATP- and GTP-binding proteins. *Trends Biochem Sci*. 1990;15(11):430-434.
48. Geer LY, Domrachev M, Lipman DJ, Bryant SH. CDART: protein homology by domain architecture. *Genome Res*. 2002;12(10):1619-1623.
49. Sievers F, Wilm A, Dineen D. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. *Mol Syst Biol*. 2011;7:539.
50. Moreno-Del Álamo M, Tabone M, Muñoz-Martínez J, Valverde JR, Alonso JC. Toxin ζ reduces the ATP and modulates the uridine diphosphate-N-acetylglucosamine pool. *Toxins*. 2019;11(1):29.
51. Trott O, Olson AJ. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J Comput Chem*. 2010;31(2):455-461.

52. Ivanova L, Tammiku-Taul J, García-Sosa AT, Sidorova Y, Saarma M, Karelson M. Molecular dynamics simulations of the interactions between glial cell line-derived neurotrophic factor family receptor GFRα1 and small-molecule ligands. *ACS Omega*. 2018;3(9):11407-11414.
53. Pettersen EF, Goddard TD, Huang CC. UCSF Chimera—a visualization system for exploratory research and analysis. *J Comput Chem*. 2004;25(13):1605-1612.
54. Hussain AS, Shanthi V, Sheik SS, Jeyakanthan J, Selvarani P, Sekar K. PDB Goodies—a web-based GUI to manipulate the Protein Data Bank file. *Acta Crystallogr D Biol Crystallogr*. 2002;58(Pt 8):1385-1386.
55. Ramachandran B, Jeyakanthan J, Lopes BS. Molecular docking, dynamics and free energy analyses of *Acinetobacter baumannii* OXA class enzymes with carbapenems investigating their hydrolytic mechanisms. *J Med Microbiol*. 2020;69(8):1062-1078.
56. Milani M, Pesce A, Bolognesi M, Bocedi A, Ascenzi P. Substrate channeling: molecular bases. *Biochem Mol Biol Educ*. 2003;31(4):228-233.
57. Ramachandran B, Srinivasadesikan V, Chou TM, Jeyakanthan J, Lee SL. Atomistic simulation on flavonoids derivatives as potential inhibitors of bacterial gyrase of *Staphylococcus aureus*. *J Biomol Struct Dyn*. 2020:1-14.
58. Zhu K, Borrelli KW, Greenwood JR. Docking covalent inhibitors: a parameter free approach to pose prediction and scoring. *J Chem Inf Model*. 2014;54(7):1932-1940.

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

How to cite this article: Karthika A, Ramachandran B, Chitra J, et al. Molecular dynamics simulation of Toxin-Antitoxin (TA) system in *Acinetobacter baumannii* to explore the novel mechanism for inhibition of cell wall biosynthesis: Zeta Toxin as an effective therapeutic target. *J Cell Biochem*. 2021;1-16. https://doi.org/10.1002/jcb.30137

View publication stats